Market Expectations and the Macroeconomy: Evidence from the Oil and Gas Industry

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Abstract

Measures of market expectations, such as survey data or asset price changes, are often limited to the first moment (the mean). To overcome this limitation, this paper extracts the complete, forward-looking risk-neutral density (RND) from American options on the S&P Oil & Gas Exploration & Production ETF (XOP) using the iterative implied binomial tree method of Tian (2011). First, I show that the higher moments of the energy sector's RND possess significant in-sample and out-of-sample predictive power for real economic activity. The evidence is strongest and most robust for industrial production, while the result for the unemployment rate is sizable at the baseline but more sensitive to the choice of estimation/evaluation split. Second, changes in the RND around OPEC announcements provide a novel measure of market reaction. I use these to construct an additional instrument that identifies the effects of oil supply news with greater precision than using the Känzig (2021) measure alone.

1 Introduction

Most empirical strategies for measuring market expectations recover information about the mean of some future variable. Futures prices, survey point forecasts, and high-frequency asset price changes in event studies are all first-moment objects. Yet there are strong reasons to believe the rest of the distribution matters. Uncertainty about future outcomes depresses investment and hiring (Bloom, 2009). Tail risks and disaster probabilities are priced and affect consumption dynamics (Backus et al., 2011). Central bank announcements systematically alter the full shape of option-implied distributions, not just their means (Hattori et al., 2016; Alonso et al., 2021; Rostagno et al., 2025). If the entire distribution of expectations is economically relevant, then methods that reduce expectations to a single

number are potentially discarding valuable information. Does this additional information matter quantitatively for macroeconomic analysis?

Financial instruments, particularly options, offer a powerful, market-based alternative to coarse survey measures by allowing researchers to extract the full forward-looking risk-neutral density (RND) over the future price of an underlying asset (Breeden and Litzenberger, 1978). In empirical finance, a large literature uses option-implied information to forecast asset returns and volatility (see Christoffersen et al., 2013, for a review). For macroeconomic variables, the evidence is more limited: Bakshi et al. (2011) and Faccini et al. (2019) demonstrate that forward variances and option-implied risk aversion from the S&P 500 predict real economic activity. Similarly, Mirshahi (2023) finds that the option-implied expected returns of S&P 500 constituents also significantly forecast real economic activity. Whether the full distributional information beyond the variance, including skewness, kurtosis, and tail behavior, can be systematically used to forecast the broader real economy and serve as formal instrumental variables for structural shock identification remains a largely unexplored frontier.

In this paper, I extract daily risk-neutral densities over the log-returns of the State Street SPDR S&P Oil & Gas Exploration & Production ETF (XOP) from American options on the ETF, using the iterative implied binomial tree method of Tian (2011). I then apply the resulting distributional information to two problems in applied macroeconomics: forecasting and the identification of structural shocks.

A large literature documents that oil price movements both reflect and shape the future state of the economy (Hamilton, 1983; Kilian, 2009; Baumeister and Hamilton, 2019), making oil-linked assets a natural setting in which to look for macro-relevant distributional information. I use equity options rather than commodity options on crude oil for two reasons. First, oil sector equity prices incorporate expectations about the broader economy, such as aggregate demand and discount rates, not only commodity supply and demand fundamentals. The resulting densities therefore aggregate a wider information set. Second, the participant base in equity option markets is diverse, combining institutional investors, index funds, and retail traders rather than specialized commodity traders alone.

First, I show that higher moments of the option-implied distribution carry substantial predictive power for the U.S. economy. Three risk-neutral moments (variance, skewness, and kurtosis) extracted from 30-day options jointly forecast six macroeconomic variables: industrial production, the unemployment rate, nominal and real personal consumption expenditures, core CPI, and energy CPI. This predictive content is not captured by standard market indicators. After controlling for aggregate equity market returns (S&P 500), broad market uncertainty (VIX), energy market shocks (WTI crude oil prices), and oil-specific

implied volatility (OVX), the option-implied moments remain jointly significant in sample and deliver meaningful out-of-sample forecasting gains. The gains are largest for real activity: the out-of-sample R^2_{OS} reaches 10.5% for industrial production and 10.2% for the unemployment rate at the one-month horizon, with variance and skewness the most informative predictors. A principal component analysis of 15 distributional features confirms that these gains do not depend on the specific choice of three moments as the summary of the density.

Second, I use the daily change in the full RND around OPEC announcements as a new source of identifying variation for oil supply news shocks in a local projections with instrumental variables (LP-IV) framework, following Känzig (2021). Because the high-dimensional distributional response is hard to summarize, I use its first principal component rather than any individual moment change. This avoids taking a stand on which feature of the distributional shift is most relevant. The distributional instrument captures information that a futures-based instrument may miss, such as the resolution of uncertainty or shifts in tail expectations. Adding it raises the effective F-statistic from 8.5 to 17 under heteroskedasticity-robust standard errors, and from 7.7 to 12.9 under HAC standard errors. The stronger first stage substantially narrows the confidence bands of all impulse responses. A further benefit is that having two instruments makes an overidentification test available. The test fails to reject the null of instrument compatibility at any horizon.

This paper contributes to several strands of literature. First, the extraction of risk-neutral densities from crude oil options originated with Melick and Thomas (1997), who used a mixture of lognormals to recover implied price distributions during the Persian Gulf crisis. Later studies by Horan et al. (2004) and Datta et al. (2017) extended the descriptive analysis of these distributions, while Brown et al. (2023) used WTI American options to evaluate market participants' forecast accuracy for spot prices. Closest to my analysis is Christoffersen et al. (2022), who extract the state price density from crude oil futures and options, demonstrating that its slopes correlate with U.S. economic conditions and predict asset returns. I depart from their work by shifting the object of interest from financial forecasting to macroeconomic prediction, showing that the higher-order moments of the RND forecast real economic activity out of sample.

Second, I contribute to the literature using high-frequency asset price changes around policy announcements to identify structural shocks (Kuttner, 2001; Gürkaynak et al., 2005; Gertler and Karadi, 2015; Känzig, 2021). While instruments in this tradition almost exclusively capture shifts in the mean, the full shape of market expectations often shifts around major events. For instance, Beber and Brandt (2006) find that macroeconomic

announcements systematically alter the higher-order moments of option-implied state-price densities in the U.S. Treasury market, while Haddad et al. (2025) reveal that policy announcements embed state-contingent promises that fundamentally reshape the left tail of option-implied distributions. Building on this logic, I demonstrate that changes in the full distribution around OPEC events contain important identifying information for structural shocks. This directly addresses recent concerns regarding weak instruments and sample sensitivity, such as those raised by Kilian (2024) about Känzig's futures-based instrument, by substantially increasing estimation precision and enabling formal overidentification testing.

Finally, this paper expands the use of option-implied distributions in macroeconomic analysis. Existing work uses them to quantify disaster probabilities (Backus et al., 2011; Hilscher et al., 2025), U.S. government debt devaluation risk (Hilscher et al., 2021), or biased policy beliefs (Bauer and Chernov, 2024). My approach shares a motivation with Rostagno et al. (2025), who use rate densities to disentangle monetary policy shocks through a narrative event study. I go further by using distributional changes as a formal instrument in a local projection framework, showing that they can provide identifying variation for empirical macroeconomics.

2 Risk-Neutral Densities from American Options

While Cox and Ross (1976) introduced the idea of risk-neutral pricing for options, Breeden and Litzenberger (1978) showed how to extract risk-neutral densities from *European* options.[1] The idea of risk-neutral pricing directly connects the price of an asset with the probability distribution of its future payoff and the risk-free rate.

In the absence of arbitrage, the fundamental theorem of asset pricing guarantees the existence of a positive stochastic discount factor that prices all traded assets. Consider a generic asset with price P_t and random payoff X_{t+1}. Let $SDF_{t,t+1}$ denote the stochastic discount factor between period t and $t+1$, and let $f(x)$ denote the physical probability density of the payoff. We have

$$P_t = \mathrm{E}[SDF_{t,t+1}\, X_{t+1}] = \int SDF_{t,t+1}(x)\, x\, f(x)\, dx,$$

where the integral is taken over all possible realizations of the payoff. Define the risk-neutral density as $q(x) = SDF_{t,t+1}(x)\, f(x) / \mathrm{E}[SDF_{t,t+1}]$. Using the fact that the gross risk-free rate satisfies $R_{t,t+1} = \mathrm{E}[SDF_{t,t+1}]^{-1}$, we can rewrite the pricing equation as the

[1] See Figlewski (2018) for a review about RND estimation.

discounted expected value of the payoff under the risk-neutral measure:

$$P_t = R_{t,t+1}^{-1}\,\tilde{\mathrm{E}}\left[X_{t+1}\right] = R_{t,t+1}^{-1}\int x\,q(x)\,dx,$$

where $\tilde{\mathrm{E}}$ denotes the expectation taken under the risk-neutral density.

Applying this to an option on an underlying asset with price S_t and maturity T, consider the current price $C_t(T,K)$ of a European call option with strike price K:

$$C_t(T,K) = R_{t,T}^{-1}\,\tilde{\mathrm{E}}[\max\{S_T - K,\,0\}] = R_{t,T}^{-1}\int_K^{\infty}(s-K)\,q(s)\,ds,$$

where $q(s)$ is now the risk-neutral density over the future asset price S_T. Breeden and Litzenberger (1978) showed that this density can be extracted in a model-free way by differentiating twice with respect to the strike price:

$$q(K) = R_{t,T}\,\frac{\partial^2 C_t}{\partial K^2}. \tag{1}$$

Thus, given a cross-section of European option prices at different strikes, one can recover the risk-neutral density.

However, U.S. equity options, including options on the XOP ETF used in this paper, are generally *American* options, which can be exercised before maturity. The value of an American option can be decomposed into the value of the corresponding European option plus an early exercise premium reflecting the additional value from the right to exercise before maturity. Because of this early exercise premium, the Breeden-Litzenberger result does not apply directly, and a large part of the literature either restricts attention to European options or relies on parametric models for the risk-neutral density.

In this paper, I use the iterative implied binomial tree methodology developed by Tian (2011) for extracting risk-neutral densities and their moments from American option prices. The idea is the following: an initial estimate of European option prices is calibrated into an implied binomial tree to compute early exercise premiums. These premiums are subtracted from observed American market prices to obtain a refined European price estimate, which is then used to re-calibrate the tree. The procedure is repeated until the early exercise premiums converge, yielding a clean set of European option prices implicit in the American option market prices.

Importantly, this procedure is built on the standard implied binomial tree method of Jackwerth and Rubinstein (1996) and is not based on a parametric model such as the Barone-Adesi and Whaley (1987) approximation for dealing with the early exercise

premium, which relies on the Black-Scholes model (Black and Scholes, 1973). Moreover, it is more precise than the Cox et al. (1979) binomial tree model. Tian (2011) shows numerically that this method consistently outperforms both the Cox et al. (1979) binomial approach and the Barone-Adesi and Whaley (1987) analytical approximation, particularly for in-the-money options where the latter two methods may fail entirely. This robustness across the full range of moneyness allows one to use a broader set of option contracts without discarding in-the-money observations.

2.1 Estimation Procedure

I implement the iterative implied binomial tree method of Tian (2011) with a reformulation of the objective function and two additional modifications. The original formulation in Tian (2011) treats density smoothness as the primary objective and penalizes deviations from observed option prices. I reverse this structure: the primary objective is to minimize pricing errors, and smoothness enters as a regularization penalty weighted by λ. This is analogous to penalized least squares regression, where the pricing errors play the role of the residuals and the smoothness penalty acts as the regularizer.[2]

In addition, I make two further modifications. First, I penalize *relative* rather than absolute pricing errors, normalizing each option's mispricing by its market price. This prevents cheap options, such as deep out-of-the-money contracts, from receiving negligible weight in the objective, and ensures that the fit is balanced across all moneyness levels. Second, I augment the smoothness penalty with a first-order difference term in addition to the standard second-order term. The additional first-order term discourages large slopes in the tails of the distribution, helping the density taper smoothly to zero at the boundaries of the grid and preventing the optimizer from allocating considerable discrete mass at the endpoints of the distribution, which can arise because we are discretizing the infinite support of log-prices up to a finite boundary.

The implied binomial tree is constructed on an evenly spaced grid of $n+1$ ending log-prices. Let P_j denote the ending nodal probability at node j, i.e., the risk-neutral probability of reaching asset (log) price $S(j)$ at maturity, and let $\mathbf{P} = (P_0, P_1, \ldots, P_n)^\top$ denote the vector of all ending nodal probabilities. For a European option i with strike price K_i and payoff function $f(\cdot)$, the model price implied by the nodal probabilities is

$$V_i^{\text{model}}(\mathbf{P}) = R_{t,T}^{-1} \sum_{j=0}^{n} P_j \, f(S(j), K_i) \tag{2}$$

[2]This reparametrization is an equivalent reformulation: both formulations trace out the same set of solutions as the trade-off parameter varies.

I recover the nodal probabilities by solving the following quadratic program:

$$\min_{\mathbf{P}} \quad \lambda\, \mathbf{P}^\top (D_2 + D_1)\, \mathbf{P} \;+\; \sum_{i=1}^{m} \left(\frac{V_i^{\text{model}}(\mathbf{P}) - V_i^{\text{mkt}}}{V_i^{\text{mkt}}} \right)^2 \tag{3}$$

subject to

$$\sum_{j=0}^{n} P_j = 1 \tag{4}$$

$$S_t - PVD(t,T) = R_{t,T}^{-1} \sum_{j=0}^{n} P_j\, S(j) \tag{5}$$

$$P_j \geq 0 \quad \text{for all } j \tag{6}$$

where V_i^{mkt} is the observed market price of option i, m is the number of traded options, and λ is a smoothing parameter. The matrix $D_2 = L_2^\top L_2$ is the second-order penalty matrix, identical to the smoothness penalty in Tian (2011):

$$[L_2 \mathbf{P}]_j = P_{j-1} - 2P_j + P_{j+1} \tag{7}$$

where L_2 is the second-difference operator with entries $(1, -2, 1)$. The matrix $D_1 = L_1^\top L_1$ is the analogous first-order penalty matrix:

$$[L_1 \mathbf{P}]_j = P_{j+1} - P_j \tag{8}$$

where L_1 is the first-difference operator with entries $(-1, 1)$. For simplicity, the same smoothing parameter λ is applied to both penalty terms. Constraint (4) ensures the probabilities sum to one, constraint (5) ensures the tree correctly prices the underlying asset (with the present value of dividends as described in Section 3), and constraint (6) rules out negative probabilities.

The smoothing parameter λ governs the trade-off between fitting observed prices and producing a smooth density. Because the estimation is performed daily over a sample spanning many years, the number and liquidity of available options varies considerably across dates. I therefore select λ in a data-dependent way, scaling it proportionally to the number of option contracts m available on each date, so that the smoothness penalty remains balanced against the pricing-error term as the cross-section of options changes.[3]

[3] Tian (2011) reports that the numerical performance of the implied binomial tree is quite robust over a wide range of values for the trade-off parameter (between 0.0001 and 0.1 in his notation). Further details on

When all options are European, $V_i^{\text{model}}(\mathbf{P})$ is linear in $\mathbf{P}$ and the problem (3) is a standard quadratic program. For American options, however, this linearity breaks down because the early exercise decision depends on the probabilities themselves. Following the procedure in Tian (2011), the problem is solved iteratively:

1. Initialize the target prices as the observed American option prices: $V_i^{(0)} = V_i^{\text{mkt}}$.
2. Solve the quadratic program (3) using $V_i^{(k)}$ as the target prices to obtain the nodal probabilities $\mathbf{P}^{(k)}$.
3. Build the implied binomial tree from $\mathbf{P}^{(k)}$ and price both American and European versions of each option on the tree. Compute the early exercise premium as $EEP_i^{(k)} = V_i^{\text{Am, tree}} - V_i^{\text{Eu, tree}}$.
4. Update the target prices: $V_i^{(k+1)} = V_i^{\text{mkt}} - EEP_i^{(k)}$.
5. Repeat steps 2–4 until $|EEP_i^{(k)} - EEP_i^{(k-1)}| < \epsilon$ for all i, where ϵ is a convergence tolerance.

At convergence, the nodal probabilities $\mathbf{P}$ are consistent with the European option prices implicit in the observed American option market. Any statistic of the risk-neutral distribution can then be computed directly from the discrete distribution defined by the ending nodal probabilities. For instance, the k-th moment of the log return is $\mu_k = \sum_{j=0}^{n} P_j R_j^k$, where $R_j = \log(S(j)/S_t)$ is the log return at node j, and quantiles are obtained from the corresponding discrete cumulative distribution function.

The next section describes the data and the specific options used for the forecasting exercise.

3 Data Description

I use options on the State Street SPDR S&P Oil & Gas Exploration & Production ETF (XOP), which tracks the S&P Oil & Gas Exploration & Production Select Industry Index, a modified equal-weighted index of U.S. oil and gas exploration, production, and refining companies. XOP is the second most liquid unleveraged energy ETF, typically trading an average of approximately 5 million shares daily. The most liquid energy ETF, XLE, is heavily concentrated in Exxon Mobil and Chevron, which together account for about 40% of its portfolio weight. XOP's modified equal-weight methodology makes it more

the scaling rule and automatic diagnostics for λ are provided in Appendix A.

Table 1: Summary Statistics for XOP Options Data

Panel A: Full Sample		
Sample period	2008/01 – 2021/12	
Total observations	545,873	
Mean days to maturity	88.1	
Median days to maturity	38.0	
Mean mid price ($)	7.60	
Mean bid-ask spread ($)	0.19	
Mean daily volume	300.2	
Mean options per day	157.3	
Panel B: By Option Type		
	Calls	Puts
Observations	277,435	268,438
Mean mid price ($)	7.29	7.91
Mean bid-ask spread ($)	0.21	0.18
Mean daily volume	232.7	369.9
Mean options per day	80.2	78.6

Notes: Summary statistics for the daily XOP ETF options from January 2008 to December 2021, obtained from OptionMetrics. Panel A presents aggregate statistics for the full sample, while Panel B provides a breakdown by option type (Calls vs. Puts). Mid prices and bid-ask spreads are reported in U.S. dollars. Daily volume represents the average number of contracts traded per underlying instrument.

representative of broad oil and gas sector risk. The sample covers the period from 2008 to 2021. Although XOP was launched in 2006, its option market did not become sufficiently liquid until 2008.

Options data, closing prices, forward prices, projected dividends, and risk-free rates for XOP are all obtained from *OptionMetrics*.[4] The forward prices reported by OptionMetrics are implied from put-call parity across all available strike pairs for each date and expiration.

Table 1 reports summary statistics for the XOP options sample. The dataset contains 545,873 option observations over the period 2008 to 2021, with an average of 157 options available per trading day. The sample is roughly balanced between calls and puts. As shown in Table 10 in the Appendix, the number of listed options and trading volume grew substantially over the sample period, from approximately 1,500 observations in 2008 to nearly 100,000 in 2021, reflecting the maturation of the XOP option market.

[4]The risk-free rates come from the OptionMetrics zero-coupon yield curve, which is derived from LIBOR deposit rates and Eurodollar futures settlement prices. OptionMetrics provides zero-coupon rates at a discrete set of maturities for each trading date. Since forward prices may correspond to maturities that do not exactly coincide with the available term structure nodes, I linearly interpolate the zero-coupon rate for each required maturity using the surrounding available maturities on the same date.

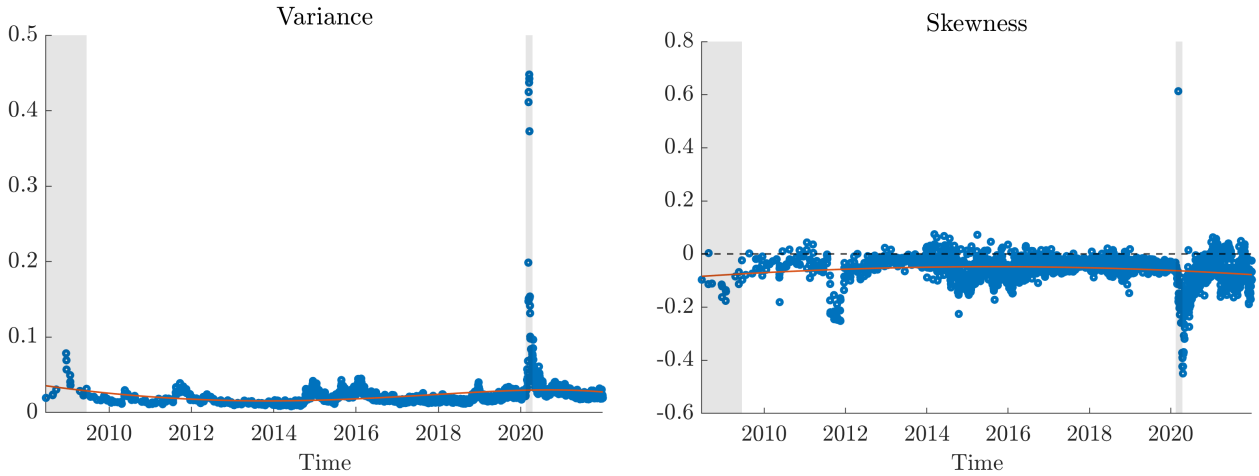


Figure 1: Risk-neutral variance and skewness from 30-day XOP options

Notes: The left panel plots the daily risk-neutral variance and the right panel plots the daily risk-neutral skewness extracted from 30-day XOP options over the sample period 2008–2021. The solid line is a cubic trend. Shaded areas indicate recession dates.

4 Macroeconomic Forecasting with Risk-Neutral Distributions

I use RNDs estimated from 30-day maturity options for my empirical exercise. To increase the number of observations, I also include options whose maturity falls within ± 3 days of the 30-day target (i.e., maturities between 27 and 33 calendar days).

This choice is motivated by data availability: 30-day options provide the densest daily coverage among available maturities. Longer maturities offer substantially fewer trading days, reducing the quality of the monthly averaging and limiting the sample available for out-of-sample evaluation. The 30-day horizon also aligns with the construction of the VIX and OVX volatility indexes.

Figure 1 plots the daily time series of the risk-neutral variance (left panel) and skewness (right panel) extracted from 30-day XOP options over the full sample. Risk-neutral variance spikes sharply at the onset of the pandemic and remains above its pre-2020 level through much of 2021, reflecting persistent uncertainty in energy markets. Risk-neutral skewness is mildly negative on average. During normal periods, it fluctuates in a narrow band around zero. It turns deeply negative during the COVID-19 episode, indicating that options markets priced in a substantially higher probability of large declines in oil and gas sector returns relative to comparably sized increases.

Figure 2 displays the estimated risk-neutral densities across the full sample. The upper panel plots densities as a function of the log price at maturity, while the lower panel plots

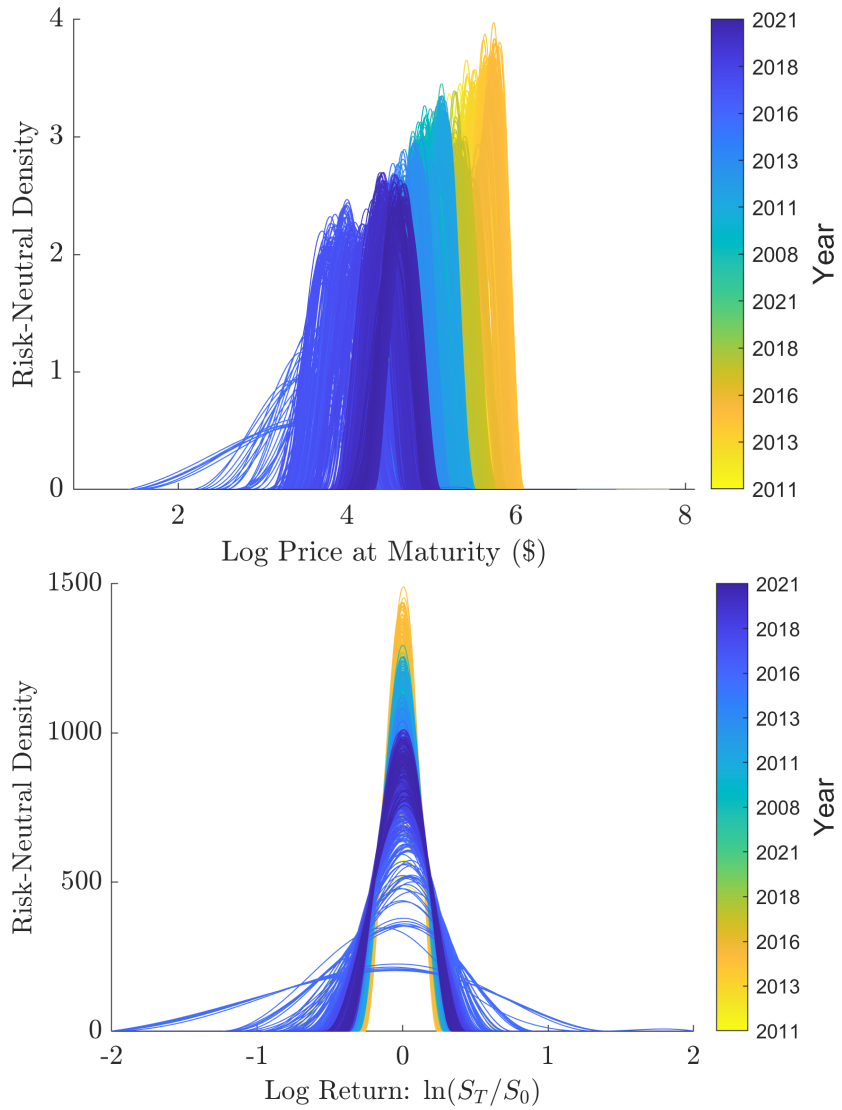


Figure 2: Evolution of risk-neutral densities over time

Notes: The upper panel plots the estimated risk-neutral densities as a function of the log price at maturity, and the lower panel plots them as a function of the log return $\ln(S_T/S_0)$, which centers all densities at zero. The color gradient represents the passage of time, with earlier dates in yellow and later dates in blue.

them as a function of the log return $\ln(S_T/S_0)$, which centers all densities at zero and makes them directly comparable across dates. While densities are tightly concentrated around the current price during tranquil periods, they exhibit substantially heavier tails and greater dispersion during the onset of the COVID-19 pandemic.

I now turn to investigating the predictive content of these distributions for real and nominal macroeconomic variables.

The forward-looking nature of risk-neutral distributions makes them natural candidates for forecasting macroeconomic outcomes. Unlike backward-looking indicators, RND

moments summarize market participants' expectation about future states of the world as embedded in options prices. The energy sector is particularly relevant in this context because of its well-documented importance (especially regarding oil prices) for aggregate economic activity (see, e.g., Hamilton, 1983; Kilian, 2009; Baumeister and Kilian, 2016; Känzig, 2021).

Risk-neutral moments can transmit predictive information through multiple channels. Under the risk-neutral measure, option-implied densities reflect both expectations about the future level of the underlying asset and compensation for bearing risk via the stochastic discount factor. In principle, the direction of the effect on a given macroeconomic variable is ambiguous: an increase in risk-neutral variance, for instance, could signal either elevated expected volatility of oil prices or a rise in risk premia, each with potentially different macroeconomic implications.

Because XOP options only gained significant liquidity starting in late 2010 (see Table 10 in Appendix B), the macro forecasting sample covers the period from October 2010 to December 2021. I focus on six macroeconomic dependent variables spanning both real activity and prices: industrial production (IP), nominal personal consumption expenditure (PCE), real personal consumption expenditure (PCEC96), the unemployment rate (UNRATE), core CPI (CPILFESL), and energy CPI (CPIENGSL).

All variables are transformed into monthly growth rates via log differences and standardized to zero mean and unit variance. The only exception is the unemployment rate, which is transformed via first differences before standardization. Four own lags are included in all specifications. This choice balances the number of regressors that can be precisely estimated given the sample size against the need to capture persistence in monthly macroeconomic data.[5]

The COVID-19 pandemic introduced extreme outliers in early 2020 that could distort inference. To address this, all results are reported for both a pre-COVID sample (through February 2020) and the full sample (through December 2021). The pre-COVID sample serves as the main specification throughout the paper. In the full-sample specifications, I include a dummy for March–April 2020 and a separate indicator for the post-April 2020 period to absorb the sharp contraction and subsequent recovery.

[5] BIC computed on univariate autoregressions favors parsimonious specifications, between 1 and 4 lags, for most variables. A notable exception is core CPI, for which BIC selects AR(12). Using AR(4) uniformly avoids ad hoc variation in lag structure across equations.

The main regression takes the form

$$
\begin{aligned}
y_{i,t} = \alpha + \beta_1 \, \text{Variance}_{t-1} + \beta_2 \, \text{Skewness}_{t-1} \\
+ \beta_3 \, \text{Kurtosis}_{t-1} + \sum_{j=1}^{4} \gamma_j \, y_{i,t-j} + \delta' \, \mathbf{X}_{t-1} + u_{i,t},
\end{aligned}
\tag{9}
$$

where $y_{i,t}$ denotes one of the six macroeconomic variables. The primary predictors are the variance, skewness, and kurtosis computed from the risk-neutral log-return distribution. These higher moments capture distributional features of expectations beyond the first moment, i.e. the return or price of an asset. To ensure consistency with the monthly frequency of the macroeconomic data, all daily financial variables and RND moments are averaged over the month. To account for potential serial correlation and heteroskedasticity in the residuals, all standard errors are computed using the Newey and West (1987) estimator with a bandwidth of 12 lags.

The control variables, $\mathbf{X}_{t-1}$, are included to ensure that the RND moments provide incremental information beyond standard financial indicators. Specifically, I control for aggregate equity market returns (S&P 500 log-differences), broad market uncertainty (the log-level of the VIX), energy market shocks (WTI crude oil log-differences), and oil-specific implied volatility (the log-level of the OVX). If the RND moments remain jointly significant after conditioning on these controls, it suggests they contain unique predictive content not captured by traditional measures of market sentiment or commodity prices.

Given the high correlations among the RND moments, individual coefficient estimates are often imprecise due to multicollinearity. The more informative test for in-sample predictive content is therefore the joint F-test of the null hypothesis $H_0 \colon \beta_1 = \beta_2 = \beta_3 = 0$, which is a test for the incremental predictive power of the RND moments.

4.1 In-Sample Predictive Content

Table 2 reports the coefficient estimates from Equation (9) for the pre-COVID sample. Given the correlations among the three RND moments, individual coefficients are often estimated imprecisely, and it is the joint pattern rather than any single coefficient that is informative. The main exception is industrial production, where the coefficient on variance (-0.620) is large and highly significant in the specification with controls (Panel B).

The coefficient on variance is negative for industrial production, meaning that increases in risk-neutral uncertainty are associated with a subsequent contraction in real activity. This is consistent with an *uncertainty channel*: higher risk-neutral variance signals greater perceived uncertainty about future energy costs, which can depress real activity as firms

Table 2: In-Sample Coefficient Estimates for Risk-Neutral Distribution Moments, Pre-COVID Sample

	IP (1)	PCE (2)	PCEC96 (3)	UNRATE (4)	Core CPI (5)	Energy CPI (6)
Panel A: Pre-COVID sample, without controls						
L.Variance	-0.677^{***}	-0.065	-0.010	0.225^{***}	0.456^{***}	-0.630
	(0.113)	(0.073)	(0.060)	(0.047)	(0.132)	(0.421)
L.Skewness	-0.147^{**}	0.060^{*}	0.014	0.085^{***}	0.153^{**}	0.350^{*}
	(0.067)	(0.032)	(0.040)	(0.024)	(0.073)	(0.205)
L.Kurtosis	0.069	0.071^{*}	-0.008	0.027	0.097	0.657^{*}
	(0.080)	(0.043)	(0.047)	(0.033)	(0.099)	(0.344)
Controls	No	No	No	No	No	No
COVID dummies	No	No	No	No	No	No
Observations	109	109	109	109	109	109
Panel B: Pre-COVID sample, with controls						
L.Variance	-0.620^{***}	-0.059	0.137	0.113^{**}	0.396^{*}	-1.571^{***}
	(0.167)	(0.094)	(0.101)	(0.056)	(0.224)	(0.513)
L.Skewness	-0.123	0.007	-0.044	0.078^{**}	0.220^{**}	0.279
	(0.085)	(0.033)	(0.042)	(0.032)	(0.085)	(0.209)
L.Kurtosis	0.085	0.013	-0.054	0.040	0.124	0.563^{*}
	(0.095)	(0.043)	(0.054)	(0.034)	(0.103)	(0.316)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
COVID dummies	No	No	No	No	No	No
Observations	109	109	109	109	109	109

Notes: Each column reports estimates from a regression of the dependent variable on the three lagged RND moments, four own lags, and additional controls (S&P 500, VIX, Oil Price, and OVX). Newey-West standard errors with 12 lags are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

delay investment and hiring (e.g., Bloom, 2009). This result is in line with Bakshi et al. (2011), who find a negative and significant coefficient for the in-sample prediction of real economic activity using forward variances extracted from S&P 500 index options, and with the contemporaneous correlations documented by Christoffersen et al. (2022), who find that the tails of the crude oil state price density steepen during periods of deteriorating economic activity. However, under the risk-neutral measure, the estimated variance reflects both physical uncertainty about future energy sector returns and compensation for bearing that risk. The predictive regressions estimated here cannot disentangle these two components, so the negative coefficient on variance is consistent with either a pure uncertainty channel, a risk-premium channel, or a combination of the two.[6]

[6] Bekaert et al. (2013), for example, decompose the VIX into a risk aversion and an uncertainty component.

For the unemployment rate, the variance coefficient is positive (0.113) and significant in Panel B, directionally consistent with the negative effect on output: higher energy sector uncertainty is associated with a subsequent rise in unemployment. Notably, this relationship persists even after controlling for broad market uncertainty (VIX) and oil-specific volatility (OVX), suggesting that the energy RND moments capture a distinct channel of transmission that is not merely a proxy for existing uncertainty benchmarks.

The coefficients on the two price indices display an interesting contrast. The variance coefficient is positive for core CPI (0.396) but negative for energy CPI (-1.571) in Panel B, suggesting that energy uncertainty may affect these two components of inflation through different mechanisms. However, the positive coefficient on core CPI is not stable across samples: it flips sign in the full sample (Panel D), so this pattern should not be overinterpreted.

Table 3 presents the results of the same specifications estimated on the full sample, including the COVID-19 period. The coefficient magnitudes change substantially, driven by the extreme observations of 2020. Despite accounting for this period with dummies for March–April 2020 and the post-COVID recovery, the unprecedented co-movement between risk-neutral variance and real activity inflates many of the estimates. For instance, the variance coefficient on the unemployment rate rises sharply from 0.113 to 1.029, and the coefficient for consumption roughly doubles. A notable exception is the IP equation, whose coefficients are more stable across samples: the variance coefficient changes only from -0.620 to -0.689. Consequently, these full-sample estimates should be interpreted with caution.

Table 4 reports the joint F-tests for the null that all three RND moment coefficients are zero. In the pre-COVID sample without controls (Panel A), the RND moments are jointly significant for five of the six variables: industrial production, nominal consumption, the unemployment rate , core CPI, and energy CPI. Real consumption is the only variable that is not significant without controls.

With controls (Panel B), industrial production, the unemployment rate, core CPI, and energy CPI remain significant at the 5% level or better. For these four variables, the RND moments contain predictive information beyond what is captured by standard financial indicators. Nominal and real consumption lose significance once controls are included ($F = 0.18$ and $F = 0.90$, respectively), indicating that for these variables the predictive content of the RND moments largely overlaps with the information already contained in the S&P 500, VIX, oil price, and OVX.

The risk-neutral variance extracted here similarly reflects both channels, though from sector-specific rather than aggregate options.

Table 3: In-Sample Coefficient Estimates for Risk-Neutral Distribution Moments, Full Sample

	IP (1)	PCE (2)	PCEC96 (3)	UNRATE (4)	Core CPI (5)	Energy CPI (6)
Panel C: Full sample, without controls						
L.Variance	-0.709^{***}	-0.448^{***}	-0.428^{***}	0.899^{***}	-0.044	-0.291^{***}
	(0.027)	(0.087)	(0.119)	(0.107)	(0.031)	(0.100)
L.Skewness	-0.136^{**}	-0.004	-0.063	0.353^{***}	-0.078^{**}	0.332^{**}
	(0.059)	(0.063)	(0.093)	(0.070)	(0.039)	(0.160)
L.Kurtosis	0.115	0.185^{*}	0.178	0.127	-0.081	0.298^{*}
	(0.082)	(0.109)	(0.160)	(0.105)	(0.061)	(0.165)
Controls	No	No	No	No	No	No
COVID dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	131	131	131	131	131	131
Panel D: Full sample, with controls						
L.Variance	-0.689^{***}	-0.490^{***}	-0.477^{***}	1.029^{***}	-0.158^{*}	-0.194
	(0.033)	(0.070)	(0.111)	(0.098)	(0.085)	(0.167)
L.Skewness	-0.102	0.055	0.029	0.173^{**}	-0.063	0.436^{***}
	(0.091)	(0.093)	(0.142)	(0.077)	(0.089)	(0.153)
L.Kurtosis	0.093	0.212^{*}	0.234	-0.017	-0.045	0.353^{**}
	(0.088)	(0.113)	(0.173)	(0.107)	(0.066)	(0.156)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
COVID dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	131	131	131	131	131	131

Notes: Each column reports estimates from a regression of the dependent variable on the three lagged RND moments, four own lags, and additional controls (S&P 500, VIX, Oil Price, and OVX). COVID dummies include a March–April 2020 indicator and a post-April 2020 indicator. Newey-West standard errors with 12 lags are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

In the full sample, the F-statistics for real activity variables increase considerably (e.g., from 6.37 to 210.93 for industrial production, from 3.32 to 99.73 for unemployment), but this increase is driven by the COVID episode rather than by genuinely stronger predictive power. For the price indices, the full sample with controls weakens the evidence somewhat: core CPI retains significance ($F = 3.45$, $p = 0.019$) but energy CPI is only marginally significant ($F = 2.84$, $p = 0.041$), consistent with the volatile energy price swings during the pandemic obscuring the more stable pre-COVID relationship.

As a robustness check, I augment the baseline control set with a measure of credit market conditions: the Gilchrist and Zakrajšek (2012) excess bond premium. Table 11 (Appendix B) reports the results. Focusing on the pre-COVID sample, the two variables for which the baseline results are strongest remain significant: industrial production ($F = 8.37$,

Table 4: Joint Significance of Risk-Neutral Distribution Moments

	IP (1)	PCE (2)	PCEC96 (3)	UNRATE (4)	Core CPI (5)	Energy CPI (6)
Panel A: Pre-COVID sample, without controls						
F-statistic	16.52***	4.47***	1.72	10.95***	3.99***	3.85**
	(0.000)	(0.005)	(0.167)	(0.000)	(0.010)	(0.012)
Panel B: Pre-COVID sample, with controls						
F-statistic	6.37***	0.18	0.90	3.32**	3.15**	3.13**
	(0.001)	(0.910)	(0.443)	(0.023)	(0.029)	(0.029)
Panel C: Full sample, without controls						
F-statistic	361.97***	12.98***	5.80***	47.07***	5.78***	6.48***
	(0.000)	(0.000)	(0.001)	(0.000)	(0.001)	(0.000)
Panel D: Full sample, with controls						
F-statistic	210.93***	25.09***	9.97***	99.73***	3.45**	2.84**
	(0.000)	(0.000)	(0.000)	(0.000)	(0.019)	(0.041)

Notes. The table reports F-statistics for the joint null hypothesis $H_0 : \beta_1 = \beta_2 = \beta_3 = 0$ from the regressions specified in (9). p-values are reported in parentheses. The test evaluates the incremental predictive content of the RND moments beyond control variables and COVID-19 indicator variables (Panels C and D). Panel B represents the baseline specification. Newey-West standard errors with 12 lags are employed to adjust for heteroskedasticity and autocorrelation. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

$p < 0.001$) and the unemployment rate ($F = 4.55$, $p = 0.005$). The F-statistics for IP and the unemployment rate are, in fact, slightly larger than in the baseline, indicating that the RND moments and credit conditions capture distinct sources of variation in future real economic activity. As before, the tests fail to find additional predictive ability for real and nominal consumption growth. Core and Energy CPI remain significant at the 5% and 10% levels, respectively. Overall, the predictive content of the RND moments for real activity is not driven by credit market conditions.

4.2 Out-of-Sample Forecasting

The in-sample results in the previous section establish that RND moments contain statistically significant predictive information for several macroeconomic variables. However, in-sample significance can be misleading if the estimated relationships are unstable or driven by a few influential observations, and the in-sample predictive content of financial variables is often a poor guide to out-of-sample forecast performance (Stock and Watson, 2003). Out-of-sample evaluation provides a more demanding test: the model must produce

forecasts using only data available at the time of the forecast.

To perform the out-of-sample analysis, I use a recursive (expanding-window) scheme. At each date t in the evaluation period, I estimate the full model and the AR benchmark using all observations from the beginning of the sample through $t-1$, then form a one-step-ahead forecast for date t. The initial estimation window contains 70 observations, and the first forecast is formed in December 2016. The evaluation period ends in February 2020, yielding $n = 39$ out-of-sample forecasts. As in the in-sample analysis, both the full model and the AR benchmark include four own lags and financial controls.

I assess out-of-sample performance using two statistics. The first is the out-of-sample R^2 of Campbell and Thompson (2008), defined as

$$R^2_{OS} = 1 - \frac{\sum_t (y_t - \hat{y}_t^{\text{full}})^2}{\sum_t (y_t - \hat{y}_t^{\text{AR}})^2}, \tag{10}$$

where $\hat{y}_t^{\text{full}}$ and $\hat{y}_t^{\text{AR}}$ are the forecasts from the full model and the AR benchmark, respectively. A positive R^2_{OS} indicates that the full model produces lower mean squared forecast errors than the benchmark.

The second is the MSPE-adjusted statistic of Clark and West (2007), designed for comparing nested models out of sample. The Clark-West adjustment accounts for the fact that, under the null of equal predictive ability, the larger model estimates additional parameters that are zero in population, introducing noise that biases the raw MSPE comparison against it. The adjusted series is

$$\hat{f}_t = (y_t - \hat{y}_t^{\text{AR}})^2 - \left[(y_t - \hat{y}_t^{\text{full}})^2 - (\hat{y}_t^{\text{AR}} - \hat{y}_t^{\text{full}})^2 \right], \tag{11}$$

and the test statistic is the t-statistic from a regression of $\hat{f}_t$ on a constant, computed with Newey-West standard errors.[7] Following the Monte Carlo evidence in Clark and West (2007), I use standard normal critical values for the one-sided test.

The evaluation is restricted to the pre-COVID period. Including the COVID episode distorts the R^2_{OS} in both directions. For real activity variables, the AR benchmark produces large forecast errors during the unprecedented contraction and recovery of 2020, mechanically inflating R^2_{OS} to large levels even though the Clark-West statistics are only marginally significant. For the price indices, the pattern reverses abd the full model's additional parameters amplify forecast errors during the volatile energy price swings of

[7] Given the small out-of-sample window, I use one lag in the Newey-West variance estimator. This is the conservative choice for the Clark-West test: increasing the number of lags in this sample produces *smaller* standard errors.

Table 5: Out-of-Sample Forecasting Results: Three Standardized Moments

	IP (1)	PCE (2)	PCEC96 (3)	UNRATE (4)	Core CPI (5)	Energy CPI (6)
R^2_{OS} (%)	10.46	−10.51	−0.95	10.19	1.44	−2.18
CW	2.43***	−1.58	0.18	2.13**	0.80	1.21

Notes: Out-of-sample forecasts are generated using an expanding window with evaluation through February 2020 ($n = 39$ forecast observations). Both the full model and the AR benchmark include four own lags and financial controls (*L*.S&P 500, *L*.VIX, *L*.Oil Price, *L*.OVX). $R^2_{OS} = 1 - \text{MSFE}_{\text{Full}}/\text{MSFE}_{\text{AR}}$ following Campbell and Thompson (2008). CW is the MSPE-adjusted statistic of Clark and West (2007) for the null hypothesis that the MSFE of the AR benchmark is less than or equal to the MSFE of the full model (one-sided test); * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ based on standard normal critical values.

2020.

Two caveats about the out-of-sample evaluation deserve emphasis. First, the evaluation window contains only $n = 39$ monthly observations. With so few observations, the R^2_{OS} statistic can be sensitive to a small number of forecasts. The Clark-West statistic, which corrects for the finite-sample penalty of estimating additional parameters, provides more reliable inference in this setting.[8] Second, the location of the split between estimation and evaluation is not innocuous. The full model estimates 12 parameters (a constant, three RND moments, four own lags, and four financial controls), so even the baseline 70-observation estimation window provides only roughly six observations per parameter. At the shorter estimation windows I examine in the robustness exercise below, this ratio decreases even more, enough that individual regression coefficients can be driven by a handful of observations. The baseline split is therefore chosen to give the largest estimation window consistent with a non-trivial evaluation window.

Table 5 reports the out-of-sample results using the three lagged centered moments (variance, skewness, kurtosis) as additional predictors. Industrial production is the clearest success. The full model achieves an R^2_{OS} of 10.46% and the Clark-West statistic ($t = 2.43$) rejects the null of equal predictive ability at the 1% level. The unemployment rate also shows a sizable baseline improvement, with an R^2_{OS} of 10.19% and a significant Clark-West statistic ($t = 2.13$, $p = 0.017$). As discussed in the robustness analysis below, this result is more sensitive to the choice of estimation/evaluation split than the industrial production result is. Core CPI shows a positive R^2_{OS} of 1.44%, though the Clark-West statistic is not

[8] Clark and West (2007) show via simulations that the MSPE-adjusted statistic performs well even in small samples. For instance, in a setup with an initial estimation window of 80 observations and an out-of-sample period of 40 observations, similar to the 70/39 split used here, the statistic's distribution conforms well to asymptotic normality.

significant ($t = 0.80$). For energy CPI, the R^2_{OS} is negative (-2.18%) and the Clark-West statistic is positive but insignificant ($t = 1.21$). For nominal and real consumption, the full model does not improve upon the AR benchmark.

To put these numbers in context, Campbell and Thompson (2008) argue that an R^2_{OS} as low as 0.5% can be economically meaningful for mean-variance investors when predicting equity returns with monthly data. Faccini et al. (2019) use S&P 500 option-implied risk aversion to forecast U.S. real economic activity and report an R^2_{OS} of 1.9% for industrial production at the one-month horizon. The R^2_{OS} obtained here for industrial production varies from roughly 1% to 10% across estimation/evaluation splits (see below), which places the result in the same ballpark as or above this benchmark, though the comparison is not direct because of differences in sample period and benchmark specification. Finally, the AR benchmark already controls for the oil price level, so the additional predictive content of the XOP option-implied RND moments cannot be attributed to oil price information alone.

To gauge the sensitivity of these results to the split point, Appendix Table 12 reproduces the exercise for three alternative initial estimation windows (39, 50, and 60 observations), each with the estimation sample fixed to run from October 2010 to February 2020. The shortest choice, $n = 39$, is the mirror image of the baseline: it uses the baseline evaluation window for estimation and the baseline estimation window for evaluation. The industrial production result is stable: the Clark-West statistic remains significant at the 5% level or better at every alternative split, consistent with the baseline, even as the R^2_{OS} falls from the baseline value of 10.5% to a range of 1.1% to 5.6% at the alternatives. This pattern, a moving R^2_{OS} magnitude combined with a stably significant Clark-West statistic, is what one would expect if the underlying predictive relationship is stable and the variation reflects estimation noise that is largest at the shortest estimation windows. The unemployment rate result, in contrast, is specific to the baseline split: both the R^2_{OS} and the Clark-West statistic become small and statistically insignificant when the split is changed. Whether this reflects a genuinely weaker signal or a signal that requires more estimation data to separate from the noise cannot be settled with the available sample. Energy CPI shows a third pattern, with significant evidence at the shorter initial windows and weaker evidence at the baseline. The bottom line is that the forecasting content of the RND for real activity is most securely attributed to industrial production.

4.3 Exploiting the Full Risk-Neutral Distribution via Principal Components

The forecasting exercises so far have relied on three low-order moments (variance, skewness, kurtosis) as summaries of the distribution. As an alternative, I summarize the density through principal components extracted from a larger set of 15 RND features: the risk-neutral mean $\tilde{\mu}$, the risk-neutral variance $\tilde{\sigma}^2$, the 3rd through 10th risk-neutral standardized moments (defined as $\tilde{\mathrm{E}}\left[((X-\tilde{\mu})/\tilde{\sigma})^k\right]$), and five percentiles of the risk-neutral distribution (P_{10}, P_{25}, P_{50}, P_{75}, P_{90}). PCA extracts the common variation across these features without requiring a choice of which individual summaries are most informative.

Over the full pre-COVID sample, the first principal component accounts for 71.8% of the total variance, the second for 24.3%, and the third for 2.6%. I report results for $k = 1$, 2, and 3 components. For the out-of-sample exercise, both the PCA loadings and the regression coefficients are re-estimated at each step of the expanding window, so that no future information enters the forecast at any point.[9] I maintain the same specification as in the baseline: four own lags of the dependent variable and the four financial controls.

Table 6 reports the in-sample F-tests for joint significance of the principal components. With a single component, the RND information is significant only for energy CPI at the 5% level. However, once the second component is included, the F-statistics become highly significant for industrial production ($F = 11.29$, $p < 0.001$) and the unemployment rate ($F = 6.32$, $p = 0.003$), confirming that the RND contains predictive content for real activity beyond what own lags and financial controls capture. Adding a third component leaves the results broadly similar, though core CPI becomes significant with three components ($F = 2.92$, $p = 0.038$).

Table 7 reports the out-of-sample R^2_{OS} and Clark-West statistics. For industrial production, the three-moment specification achieves an R^2_{OS} of 10.46%, compared to 8.88% using three principal components. The Clark-West statistic is comparable in both cases ($t = 2.43$ and $t = 2.32$, respectively), indicating essentially equivalent predictive performance. For the unemployment rate, the three-moment specification produces a larger R^2_{OS} (10.19% versus 4.87% for the three PCs) and a higher Clark-West statistic ($t = 2.13$ versus $t = 1.58$). This sensitivity to the summarization choice parallels the sensitivity to the split choice documented above. Taken together, the comparison suggests that for the two real-activity variables where the RND moments have the strongest predictive power, the three-moment decomposition extracts the forecasting signal at least as efficiently as the data-driven PCA

[9] In the first estimation window ($n = 70$), the first three principal components explain 74.0%, 24.0%, and 2.0% of the total variance.

Table 6: In-Sample Joint Significance: Principal Components from the RND, Pre-COVID

	IP (1)	PCE (2)	PCEC96 (3)	UNRATE (4)	Core CPI (5)	Energy CPI (6)
Panel A: 1 PC						
F-statistic	0.25	0.29	1.62	1.55	0.09	4.58**
	(0.618)	(0.592)	(0.206)	(0.217)	(0.767)	(0.035)
Panel B: 2 PCs						
F-statistic	11.29***	0.14	1.37	6.32***	1.56	2.82*
	(0.000)	(0.869)	(0.259)	(0.003)	(0.215)	(0.064)
Panel C: 3 PCs						
F-statistic	8.01***	0.18	0.96	3.67**	2.92**	1.87
	(0.000)	(0.913)	(0.415)	(0.015)	(0.038)	(0.140)

Notes. The table reports F-statistics for the joint null hypothesis that all principal component coefficients equal zero. Principal components are extracted from 15 lagged RND features (ten standardized moments and five percentiles) using PCA on the correlation matrix. Pre-COVID sample (October 2010 to February 2020, $n = 112$). All regressions include four own lags and financial controls (*L*.S&P 500, *L*.VIX, *L*.Oil Price, *L*.OVX). p-values in parentheses. Newey–West standard errors with 12 lags. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

summarization.

For the remaining variables, the PCA approach performs comparably or somewhat better. Core CPI, which shows a positive but insignificant R^2_{OS} of 1.44% in the three-moment specification, reaches 2.98% with three principal components, and the Clark-West statistic strengthens to $t = 1.03$, though it remains insignificant. Energy CPI retains negative R^2_{OS} values in both specifications. Nominal and real consumption do not benefit from either summarization method.

These results confirm the main finding for industrial production: the predictive content of the energy sector RND does not depend on how the distribution is summarized. The Clark-West statistic exceeds $t = 2$ in both the three-moment specification and the two- and three-PC specifications. For the unemployment rate, the evidence is weaker under PCA than under the three-moment decomposition and, as noted in the robustness exercise above, is also sensitive to the choice of split. The three-moment specification may be preferred for industrial production because of its parsimony and interpretability.

Appendix Table 13 reports the corresponding robustness check for the principal-components specifications, using the same three alternative estimation windows as Table 12. The industrial production result is stable: the Clark-West statistic exceeds conventional

Table 7: Out-of-Sample Forecasting Results: Principal Components from the RND, Pre-COVID

	IP (1)	PCE (2)	PCEC96 (3)	UNRATE (4)	Core CPI (5)	Energy CPI (6)
Panel A: R^2_{OS} (%)						
1 PC	−1.15	−3.67	0.09	−1.09	−3.15	−6.69
2 PCs	8.69	−5.95	2.20	1.85	−1.93	−6.28
3 PCs	8.88	−6.55	1.37	4.87	2.98	−4.30
Panel B: Clark–West statistic						
1 PC	−0.76	−0.72	0.25	0.14	−0.47	−0.16
2 PCs	2.26**	−1.09	0.93	0.96	−0.13	0.74
3 PCs	2.32**	−1.20	0.61	1.58*	1.03	0.87

Notes. Principal components are extracted from 15 lagged RND features (first ten standardized moments and five percentiles) using PCA on the correlation matrix. Both the PCA loadings and the regression coefficients are re-estimated at each step of the expanding window to avoid look-ahead bias. Pre-COVID evaluation period through February 2020 ($n = 39$ forecast observations). Both the full model and the AR benchmark include four own lags and financial controls (*L*.S&P 500, *L*.VIX, *L*.Oil Price, *L*.OVX). $R^2_{OS} = 1 - \text{MSFE}_{\text{Full}}/\text{MSFE}_{\text{AR}}$ following Campbell and Thompson (2008). CW is the MSPE-adjusted statistic of Clark and West (2007) (one-sided test). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

thresholds at every alternative split in the three-PC specification and at two of three in the two-PC specification, consistent with the baseline significance in Table 7. This robustness, across both the choice of split point and the number of principal components, reinforces the industrial production finding.

The fact that two principal components, rather than one, are needed for the predictive content to emerge is itself informative: it indicates that the macro-relevant information in the RND resides in more than one dimension of distributional variation, consistent with the paper's broader argument that moving beyond the first moment of market expectations captures information not already contained in standard financial indicators.

The next section turns from forecasting to identification, using high-frequency changes in the RND around OPEC announcements as an instrument for oil supply news shocks.

5 Distributional Instruments for Oil Supply News Shock Identification

A large body of work uses high-frequency changes in asset prices around policy announcements to identify exogenous macroeconomic shocks. In the monetary policy literature, changes in federal funds or Eurodollar futures rate in narrow windows around FOMC meetings serve as instruments for monetary policy surprises (Kuttner, 2001; Gürkaynak et al., 2005; Gertler and Karadi, 2015; Nakamura and Steinsson, 2018). Känzig (2021) extends this approach to the oil market, using changes in oil futures prices around OPEC production announcements to identify oil supply news shocks. In these applications, the high-frequency instruments typically capture variation in the *mean* of an asset.

Options markets, however, price the entire risk-neutral distribution, not just its first moment. Two OPEC announcements may produce the same futures price change while generating very different distributional responses: one may resolve uncertainty about future supply while the other amplifies it. Conversely, an announcement may leave the expected price unchanged while substantially altering the variance or skewness of the risk-neutral distribution.[10] Figure 3 illustrates these two different cases. The left panel shows a pure shift in the mean of the RND, corresponding to the type of variation that futures-based instruments capture. The right panel shows a pure change in variance with no change in the mean. A futures-based instrument is uninformative in the right-panel scenario, even though the distributional shift may reflect economically meaningful news about oil supply prospects.

This distinction is not merely hypothetical. Several papers document substantial movements in option-implied distributions around oil market events (Melick and Thomas, 1997; Horan et al., 2004; Datta et al., 2017). I go further and use distributional changes as instruments in a structural framework.

I build on this observation by using high-frequency changes in the risk-neutral distribution as instruments for oil supply shock identification. The key idea is that if OPEC announcements contain information beyond shifts in the mean, then distributional changes provide an additional source of identifying variation and enable an overidentification test that is not available with a single instrument.

[10] Beber and Brandt (2006), in fact, find that policy announcements reduce the uncertainty extracted from U.S. Treasury bond futures options. Similarly, Rostagno et al. (2025) move beyond the mean by using option-implied rate densities to identify negative interest rate policy and forward guidance shocks through a narrative-based event study.

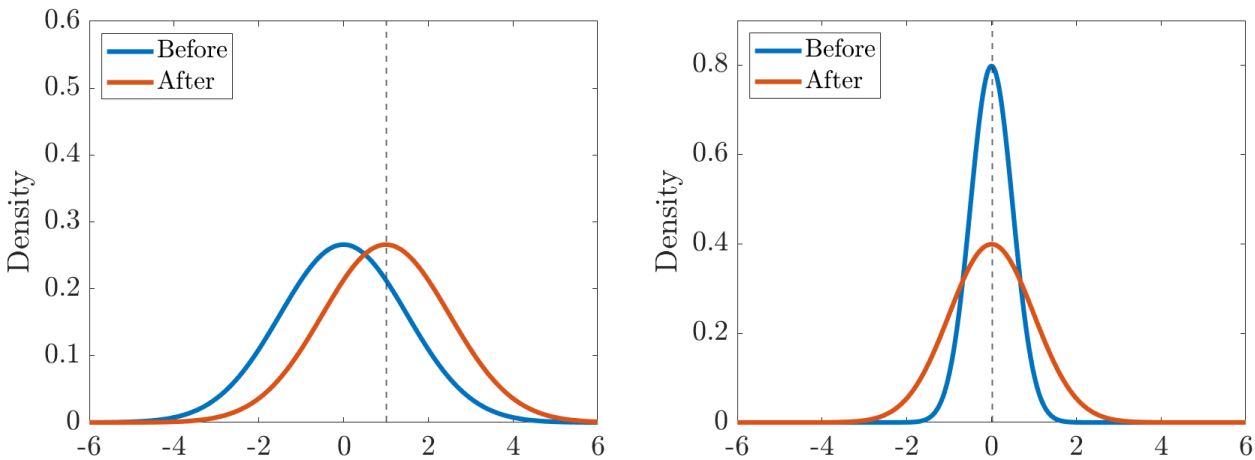


Figure 3: Risk-neutral density shifts: mean shift vs. shape change

Notes: The left panel shows a pure shift in the mean of the risk-neutral distribution. The right panel illustrates a change in the shape of the distribution with no change in the mean, representing the additional information contained in options prices.

5.1 Econometric Framework

I estimate impulse response functions for the oil supply news shock using local projections with instrumental variables (LP-IV), estimated via GMM (Stock and Watson, 2018; Jordà, 2023). The estimating equation at horizon h is

$$y_{i,t+h} = \beta_{h,i}\, p_t^{\text{oil}} + \mathbf{x}_{t-1}'\gamma_{h,i} + u_{i,t+h}, \qquad h = 0, 1, \ldots, H, \tag{12}$$

where $y_{i,t+h}$ is the level of variable i at horizon h, p_t^{oil} is the log real oil price, and $\mathbf{x}_{t-1}$ contains lagged controls. The variables used, other than the real oil price, are the same as those used by Känzig (2021): world oil production, world oil inventories, world industrial production, U.S. industrial production, and the U.S. consumer price index. The structural impulse response $\beta_{h,i}$ is identified by instrumenting p_t^{oil} with high-frequency surprises $\mathbf{z}_t$ observed around OPEC announcements.

With a single instrument, the model is exactly identified and the exclusion restriction cannot be tested. With multiple instruments, the model is overidentified and the Hansen (1982) J-test provides a diagnostic: under the null, all instruments identify the same $\beta_{h,i}$, and rejection implies that the moment conditions from different instruments are incompatible, meaning they do not isolate the same response to the identified shock.

I follow Känzig (2021) in specifying 18 lags with variable-specific controls. For the oil market variables (real oil price, world oil production, world oil inventories, world industrial production), the control set includes lags of all four oil block variables. For U.S. industrial production and U.S. CPI, the controls additionally include own lags. Following

Känzig (2021), I partial out the controls on the full estimation sample and run the IV on the proxy subsample (April 1983 to February 2020, excluding the COVID-19 pandemic). Standard errors are computed using Newey-West HAC with bandwidth $h+1$ (where h is the impulse response horizon).

The LP-IV approach is preferred over a proxy-SVAR for this application for two primary reasons. First, LP-IV does not require invertibility of the underlying VAR: therefore, identification remains valid even if the structural shocks cannot be perfectly recovered from the VAR innovations (Stock and Watson, 2018; Plagborg-Møller and Wolf, 2021). Second, overidentification tests are straightforward to implement within the LP-IV framework.

Identification in LP-IV relies on relevance, contemporaneous exogeneity, and lead-lag exogeneity conditions (Stock and Watson, 2018; Jordà, 2023). Suppose the variables in the system are driven by a vector of additive structural shocks. Let ϵ_t^1 denote the target shock of interest, namely the oil supply news shock (Känzig, 2021). Let ϵ_t^j (for $j > 1$) denote all other structural shocks. To validly identify the dynamic effects of ϵ_t^1, the vector of instruments $\mathbf{z}_t$ must satisfy the following structural assumptions:[11]

$$\mathrm{E}[\mathbf{z}_t \epsilon_t^1] \neq \mathbf{0}, \tag{13}$$

$$\mathrm{E}[\mathbf{z}_t \epsilon_t^j] = \mathbf{0} \quad \text{for all } j > 1, \tag{14}$$

$$\mathrm{E}[\mathbf{z}_t \epsilon_s^j] = \mathbf{0} \quad \text{for all } j \text{ and } s \neq t. \tag{15}$$

Equations (13), (14), and (15) represent the relevance, contemporaneous exogeneity, and lead-lag exogeneity conditions, respectively.

Under these assumptions, the instruments correctly isolate variations in the endogenous oil price variable, p_t^{oil}, that are driven solely by the target structural shock. This structural setup yields the standard moment conditions required to estimate the local projection in Equation (12):

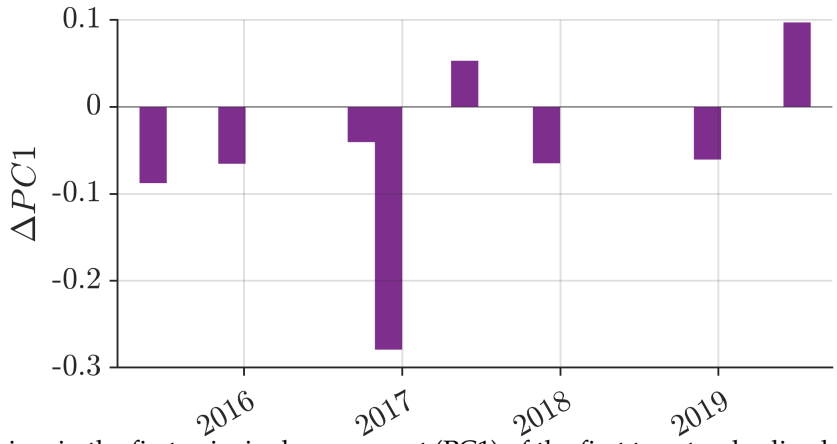
$$\mathrm{E}[\mathbf{z}_t p_t^{\text{oil}}] \neq \mathbf{0}, \tag{16}$$

$$\mathrm{E}[\mathbf{z}_t u_{i,t+h}] = \mathbf{0} \quad \text{for all } i \text{ and } h \geq 0, \tag{17}$$

where $u_{i,t+h}$ denotes the projection residual for variable i at horizon h.

[11] This assumes that the control variables in Equation (12) are included solely to improve estimation precision. If the instruments are only valid conditional on these controls, the moment conditions must hold after partialling out these observable variables.

Figure 4: Surprises in the First Principal Component around OPEC Announcements



Notes: Daily surprises in the first principal component (PC1) of the first ten standardized moments and five percentiles of the risk-neutral distribution (P_{10}, P_{25}, P_{50}, P_{75}, P_{90}) of the XOP ETF around OPEC announcements.

5.2 Instruments

The Känzig (2021) instrument $z_{K,t}$ is constructed from changes in oil futures prices around OPEC announcements and captures shifts in the conditional mean of the price distribution.[12] I supplement it with a second instrument that captures changes in the full distributional shape.

OPEC announcements can alter the distribution of expected oil sector returns in ways that go beyond shifting the mean price. A production cut announcement may resolve uncertainty about future supply, compressing the variance of the risk-neutral distribution. Alternatively, an ambiguous or contested decision may widen the distribution by increasing disagreement about future supply paths. Announcements can also shift tail expectations: a credible commitment to production discipline may reduce the probability of extreme price declines, affecting skewness and kurtosis.

I compress these distributional responses into a single index. For each OPEC announcement date identified by Känzig (2021), I compute the change in 15 RND features (10 standardized moments and 5 percentiles) from the trading day before to the announcement day and extract the first principal component, following the PCA approach in Section 4.3. Figure 4 displays the daily surprises. The daily surprises are aggregated to monthly frequency by summing within months, with zeros assigned to non-event months.

The baseline specification uses $z_{K,t}$ alongside $z_{\text{PC1},t}$, giving two excluded instruments for one endogenous variable and one overidentifying restriction. The monthly correlation

[12] For my analysis, I use the updated time series of oil futures surprises obtained from Diego Känzig's website.

Table 8: First-Stage Weak Instrument Diagnostics

	z_K		(z_K, z_{pc})	
	(1)	(2)	(3)	(4)
Effective F-statistic	7.71	8.54	12.88	17.00
Standard Errors	HAC	Robust	HAC	Robust
Observations	443	443	443	443

Notes: This table reports the first-stage Olea and Pflueger (2013) effective F-statistic, which tests for weak instruments in the presence of non-i.i.d. errors. Robust denotes standard errors that are robust strictly to heteroskedasticity. HAC denotes Heteroskedasticity and Autocorrelation Consistent (Newey-West) standard errors, using a bandwidth of 12 lags.

between the two instruments is -0.32.[13]

5.3 Results

Table 8 reports first-stage diagnostics. I evaluate instrument strength using the Olea and Pflueger (2013) effective F-statistic under both heteroskedasticity-robust and HAC standard errors with 12 lags.[14]

With the Känzig (2021) instrument alone, the effective F-statistic under robust standard errors is 8.54. This indicates a weak instrument problem in the sample ending in February 2020, consistent with Kilian (2024), who show that the explanatory power of the Känzig (2021) instrument is sensitive to the estimation period.

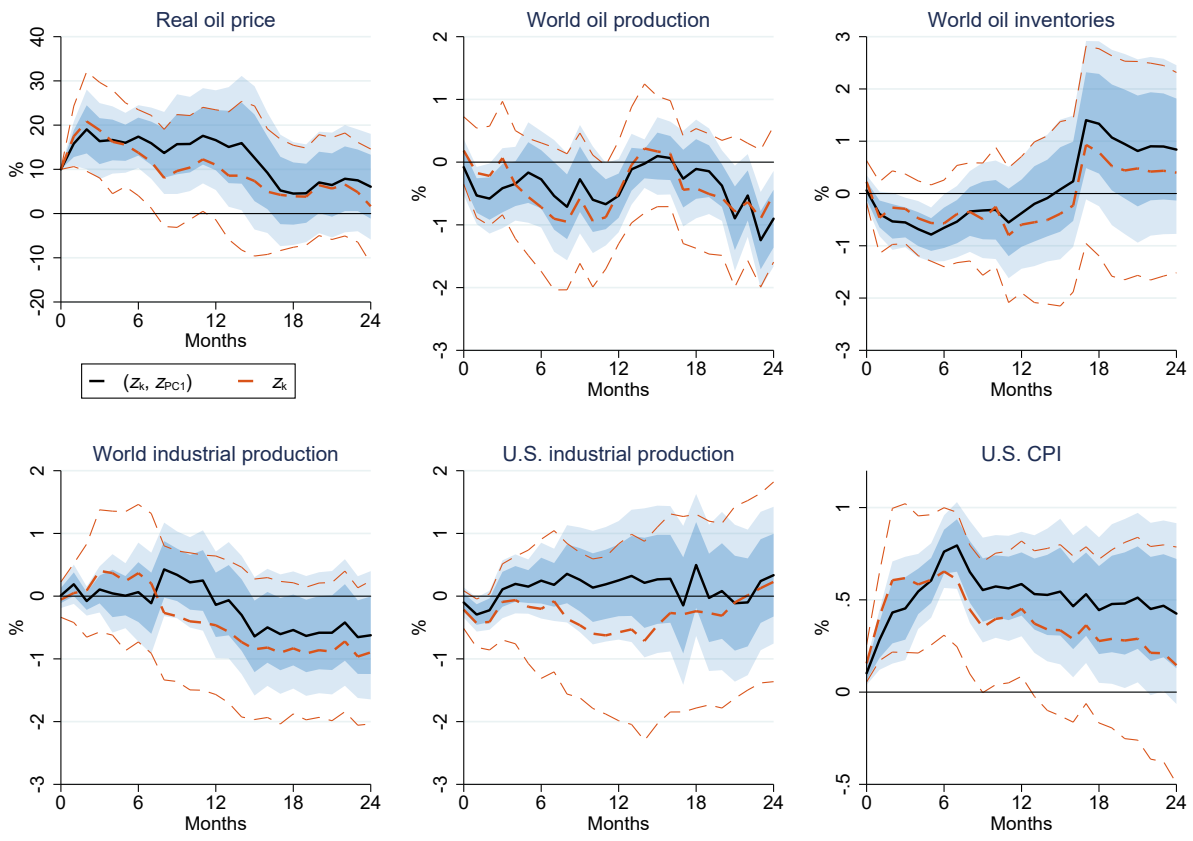
Adding $z_{\text{PC1},t}$ raises the effective F-statistic to 17.00 under robust standard errors and 12.88 under HAC standard errors. Thus, the distributional instrument considerably strengthens the first stage while providing an overidentifying restriction that enables the Hansen (1982) J-test.

Figure 5 presents impulse responses to a 10 percent increase in the real oil price for the six VAR variables. The dashed orange lines show the exactly identified specification using $z_{K,t}$ alone. The solid black lines with shaded bands show the overidentified specification using both $z_{K,t}$ and $z_{\text{PC1},t}$. Dark and light shaded regions denote 68 and 90 percent

[13] A necessary diagnostic for the lead-lag exogeneity conditions (15) is that instruments should not be forecastable by lagged macro variables. Känzig (2021) reports Granger causality tests showing that $z_{K,t}$ passes this requirement. The distributional instrument $z_{\text{PC1},t}$ similarly passes: a joint Granger causality test using 12 lags of the six VAR variables yields $F = 0.41$ ($p \approx 1$), though the result is largely mechanical since the instrument series is highly sparse, taking a value of zero in the vast majority of periods.

[14] The effective F-statistic of Olea and Pflueger (2013) is preferred over the conventional first-stage F-statistic because it remains valid under heteroskedasticity and serial correlation.

Figure 5: Impulse responses to an oil supply news shock: exact vs. overidentified specification



Notes: Impulse responses to an oil supply news shock normalized to increase the real price of oil by 10 percent on impact. The dashed orange lines correspond to the exactly identified specification using the Känzig (2021) futures surprise $z_{K,t}$ as the sole instrument. The solid black lines correspond to the overidentified specification using both $z_{K,t}$ and the distributional surprise $z_{\text{PC1},t}$. Dark and light shaded regions denote 68 and 90 percent confidence bands, respectively, constructed using Newey-West HAC standard errors with bandwidth $h+1$.

confidence bands, respectively.

The oil market variables display responses consistent with an oil supply news shock in both specifications. The real oil price increases on impact by construction and remains elevated in the horizon considered. World oil production declines gradually, and world industrial production contracts in the longer run. U.S. CPI rises persistently in the overidentified specification, reflecting the pass-through of higher energy costs. These patterns are qualitatively similar across the two specifications, but the overidentified estimates are substantially more precise: confidence bands shrink considerably across all six variables, reflecting the stronger first stage.

U.S. industrial production shows no statistically significant response in either speci-

Table 9: Hansen J-test p-values by horizon

	Oil market block				U.S. macro	
h	RPOIL	OILPROD	OILSTOCKS	WORLDIP	IP	CPI
0	—	0.221	0.262	0.593	0.337	0.225
1	0.446	0.221	0.827	0.520	0.369	0.175
2	0.626	0.229	0.359	0.697	0.240	0.193
3	0.495	0.135	0.357	0.587	0.380	0.279
4	0.941	0.971	0.466	0.481	0.388	0.839
5	0.915	0.271	0.493	0.694	0.497	0.983
6	0.377	0.323	0.797	0.505	0.414	0.551
7	0.334	0.412	0.705	0.490	0.653	0.318
8	0.125	0.554	0.981	0.129	0.170	0.185
9	0.123	0.418	0.618	0.162	0.095	0.174
10	0.133	0.385	0.878	0.212	0.075	0.228
11	0.146	0.464	0.538	0.132	0.097	0.278
12	0.145	0.880	0.603	0.449	0.176	0.379
13	0.085	0.875	0.493	0.185	0.183	0.284
14	0.104	0.672	0.443	0.290	0.116	0.236
15	0.102	0.845	0.437	0.556	0.233	0.148
16	0.099	0.849	0.425	0.358	0.399	0.175
17	0.130	0.526	0.443	0.398	0.797	0.185
18	0.185	0.392	0.415	0.218	0.246	0.203
19	0.133	0.395	0.419	0.210	0.655	0.218
20	0.184	0.660	0.410	0.195	0.389	0.237
21	0.115	0.836	0.544	0.207	0.981	0.236
22	0.115	0.801	0.428	0.172	0.846	0.224
23	0.128	0.463	0.415	0.207	0.859	0.221
24	0.171	0.399	0.412	0.188	0.882	0.225

Notes: The table reports p-values from the Hansen (1982) J-test of overidentifying restrictions at each horizon $h = 0, \ldots, 24$. The instruments are $z_{K,t}$ and $z_{\mathrm{PC1},t}$, giving one overidentifying restriction. The test statistic is computed from two-step efficient GMM with Newey-West HAC weighting (bandwidth $h + 1$). RPOIL at $h = 0$ is not tested because the impact response is normalized to 10 percent by construction.

fication, with confidence bands encompassing zero throughout the horizon. The point estimates differ across the two specifications, with the exactly identified estimate pointing slightly negative and the overidentified estimate closer to zero, but neither is distinguishable from zero. Consistent with this, the J-test reported below does not reject instrument compatibility for U.S. industrial production at any horizon, though the p-values are among the lowest across the six variables.

Table 9 reports p-values from the Hansen (1982) J-test at each horizon. The J-test does not reject at the 5 percent level for any variable at any horizon. The lowest p-value is 0.075 (U.S. industrial production, $h = 10$). Non-rejection implies that the futures-based and

distributional instruments identify statistically compatible impulse responses, consistent with both instruments capturing the same structural shock.

To interpret this result precisely, note that the J-test maintains the assumption that at least one instrument is valid, meaning it satisfies the exclusion restriction (correlated with the oil supply news shock and uncorrelated with all other structural shocks). Under this assumption, non-rejection means that the additional instrument is also consistent with the same exclusion restriction. In this application: if the Känzig (2021) futures surprise is a valid instrument for oil supply news, then the distributional instrument $z_{\text{PC1},t}$ is compatible with being valid as well.

However, the J-test cannot establish that the maintained assumption itself holds. If both instruments are contaminated by the same non-supply shocks, the test has no power to detect this. The overidentification test is informative about instrument *compatibility* but not about instrument *validity* in an absolute sense.

This observation connects to the broader debate about the interpretation of high-frequency oil market instruments. Degasperi (2026) argues that the Känzig (2021) futures surprise may mix together oil supply news with demand expectations embedded in OPEC communications. The J-test cannot resolve this concern: if OPEC announcements systematically bundle supply decisions with demand signals, and both the futures surprise and the distributional instrument respond to this bundle in similar proportions, the overidentification test will not detect the contamination. What the test does establish is that, to the extent that the distributional instrument captures a different mix of structural shocks than the futures surprise, this difference is not statistically detectable in the impulse responses.

6 Conclusion

Capturing forward-looking expectations is central to answering many questions in macroeconomics. While empirical work often relies on survey data or the first moment of asset prices, options markets provide a richer source of information by pricing the entire probability distribution of future outcomes. In this paper, I extract risk-neutral distributions from options on a broad oil and gas ETF (XOP) and demonstrate their usefulness for both macroeconomic forecasting and structural shock identification.

First, I show that the moments of the energy sector's risk-neutral distribution contain predictive power for U.S. macroeconomic variables. After controlling for aggregate stock market returns, the VIX, and the growth rate and implied volatility of crude oil prices, the RND moments remain jointly significant predictors of real economic activity. This

predictive power is strongest and most robust for industrial production, for which the RND delivers out-of-sample R^2_{OS} statistics that are significant at conventional levels across alternative choices of the estimation/evaluation split. The unemployment rate shows a similar baseline R^2_{OS} but it is sensitive to the split choice. This suggests that the shape of the option-implied distribution carries an uncertainty and tail-risk signal that standard market indicators do not.

Second, I demonstrate that these distributions are a useful tool for high-frequency identification. While the existing literature relies on changes in oil futures prices around OPEC announcements to identify oil supply news shocks, futures prices only capture shifts in the mean expectation. By constructing an instrument based on high-frequency changes in the full risk-neutral distribution, I capture shifts in market uncertainty and tail risk generated by these same announcements. Incorporating this distributional instrument in an LP-IV framework substantially strengthens the first stage, yields more precisely estimated impulse responses, and provides an overidentifying restriction that allows for the testing of instrument compatibility.

Altogether, the results in this paper highlight that moving beyond the first moment of asset prices offers substantial benefits for empirical macroeconomics. Option-implied distributions represent a market-based measure of expectations that can improve both macroeconomic forecasting and the identification of structural shocks.

References

Alonso, I., P. Serrano, and A. Vaello-Sebastià (2021). The impact of heterogeneous unconventional monetary policies on the expectations of market crashes. Working Paper 2127, Banco de España.

Backus, D., M. Chernov, and I. Martin (2011). Disasters implied by equity index options. *The Journal of Finance 66*(6), 1969–2012.

Bakshi, G., G. Panayotov, and G. Skoulakis (2011). Improving the predictability of real economic activity and asset returns with forward variances inferred from option portfolios. *Journal of Financial Economics 100*(3), 475–495.

Barone-Adesi, G. and R. E. Whaley (1987). Efficient analytic approximation of american option values. *The Journal of Finance 42*(2), 301–320.

Bauer, M. and M. Chernov (2024). Interest rate skewness and biased beliefs. *The Journal of Finance 79*(1), 173–217.

Baumeister, C. and J. D. Hamilton (2019). Structural interpretation of vector autoregressions with incomplete identification: Revisiting the role of oil supply and demand shocks. *American Economic Review 109*(5), 1873–1910.

Baumeister, C. and L. Kilian (2016). Forty years of oil price fluctuations: Why the price of oil may still surprise us. *Journal of Economic Perspectives 30*(1), 139–160.

Beber, A. and M. W. Brandt (2006). The effect of macroeconomic news on beliefs and preferences: Evidence from the options market. *Journal of monetary economics 53*(8), 1997–2039.

Bekaert, G., M. Hoerova, and M. Lo Duca (2013). Risk, uncertainty and monetary policy. *Journal of Monetary Economics 60*(7), 771–788.

Black, F. and M. Scholes (1973). The pricing of options and corporate liabilities. *Journal of Political Economy 81*(3), 637–654.

Bloom, N. (2009). The impact of uncertainty shocks. *Econometrica 77*(3), 623–685.

Breeden, D. T. and R. H. Litzenberger (1978). Prices of state-contingent claims implicit in option prices. *The Journal of Business 51*(4), 621–651.

Brown, J. P., N. Ç. Melek, J. Matschke, and S. A. Sattiraju (2023). The missing tail risk in option prices. Research Working Paper 23-02, Federal Reserve Bank of Kansas City.

Campbell, J. Y. and S. B. Thompson (2008). Predicting excess stock returns out of sample: Can anything beat the historical average? *The Review of Financial Studies 21*(4), 1509–1531.

Christoffersen, P., K. Jacobs, and B. Y. Chang (2013). Chapter 10 - forecasting with option-implied information. In G. Elliott and A. Timmermann (Eds.), *Handbook of Economic Forecasting*, Volume 2 of *Handbook of Economic Forecasting*, pp. 581–656. Elsevier.

Christoffersen, P., K. Jacobs, and X. N. Pan (2022). The state price density implied by crude oil futures and option prices. *The Review of Financial Studies 35*(2), pp. 1064–1103.

Clark, T. E. and K. D. West (2007). Approximately normal tests for equal predictive accuracy in nested models. *Journal of Econometrics 138*(1), 291–311. 50th Anniversary Econometric Institute.

Cox, J. C. and S. A. Ross (1976). The valuation of options for alternative stochastic processes. *Journal of Financial Economics 3*(1), 145–166.

Cox, J. C., S. A. Ross, and M. Rubinstein (1979). Option pricing: A simplified approach. *Journal of Financial Economics 7*(3), 229–263.

Datta, D. D., J. M. Londono, and L. J. Ross (2017). Generating options-implied probability densities to understand oil market events. *Energy Economics 64*, 440–457.

Degasperi, R. (2026). Identification of expectational shocks in the oil market using opec announcements. Temi di discussione (Economic working papers) 1516, Bank of Italy, Economic Research and International Relations Area.

Faccini, R., E. Konstantinidi, G. Skiadopoulos, and S. Sarantopoulou-Chiourea (2019). A new predictor of u.s. real economic activity: The s&p 500 option implied risk aversion. *Management Science 65*(10), pp. 4927–4949.

Figlewski, S. (2018). Risk-neutral densities: A review. *Annual Review of Financial Economics 10*(Volume 10, 2018), 329–359.

Gertler, M. and P. Karadi (2015). Monetary policy surprises, credit costs, and economic activity. *American Economic Journal: Macroeconomics 7*(1), 44–76.

Gilchrist, S. and E. Zakrajšek (2012). Credit spreads and business cycle fluctuations. *American Economic Review 102*(4), 1692–1720.

Gürkaynak, R. S., B. Sack, and E. T. Swanson (2005). Do actions speak louder than words? The response of asset prices to monetary policy actions and statements. *International Journal of Central Banking 1*(1), 55–93.

Haddad, V., A. Moreira, and T. Muir (2025). Whatever it takes? the impact of conditional policy promises. *American Economic Review 115*(1), 295–329.

Hamilton, J. D. (1983). Oil and the macroeconomy since world war ii. *J. Polit. Econ.; (United States) 91*(2), 228–248.

Hansen, L. P. (1982). Large sample properties of generalized method of moments estimators. *Econometrica 50*(4), 1029–1054.

Hattori, M., A. Schrimpf, and V. Sushko (2016). The response of tail risk perceptions to unconventional monetary policy. *American Economic Journal: Macroeconomics 8*(2), 111–136.

Hilscher, J., A. Raviv, and R. Reis (2021). Inflating Away the Public Debt? An Empirical Assessment. *The Review of Financial Studies 35*(3), 1553–1595.

Hilscher, J., A. Raviv, and R. Reis (2025). How likely is an inflation disaster? *The Review of Financial Studies 39*(3), 744–782.

Horan, S. M., J. H. Peterson, and J. Mahar (2004). Implied volatility of oil futures options surrounding opec meetings. *The Energy Journal 25*(3), 103–125.

Jackwerth, J. C. and M. Rubinstein (1996). Recovering probability distributions from option prices. *The Journal of Finance 51*(5), 1611–1631.

Jordà, Ò. (2023). Local projections for applied economics. *Annual Review of Economics 15*(1), 607–631.

Kilian, L. (2009). Not all oil price shocks are alike: Disentangling demand and supply shocks in the crude oil market. *American Economic Review 99*(3), 1053–69.

Kilian, L. (2024). How to construct monthly var proxies based on daily surprises in futures markets. *Journal of Economic Dynamics and Control 168*, 104966.

Kuttner, K. N. (2001). Monetary policy surprises and interest rates: Evidence from the fed funds futures market. *Journal of Monetary Economics 47*(3), 523–544.

Känzig, D. R. (2021). The macroeconomic effects of oil supply news: Evidence from opec announcements. *American Economic Review 111*(4), 1092–1125.

Melick, W. R. and C. P. Thomas (1997). Recovering an asset's implied pdf from option prices: An application to crude oil during the gulf crisis. *The Journal of Financial and Quantitative Analysis 32*(1), 91–115.

Mirshahi, M. (2023). *Essays on Option-Implied Measures of Risk*. Phd thesis, Queen Mary University of London.

Nakamura, E. and J. Steinsson (2018). High-frequency identification of monetary non-neutrality: The information effect. *The Quarterly Journal of Economics 133*(3), 1283–1330.

Newey, W. K. and K. D. West (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica 55*(3), 703–708.

Olea, J. L. M. and C. Pflueger (2013). A robust test for weak instruments. *Journal of business & economic statistics 31*(3), 358–369.

Plagborg-Møller, M. and C. K. Wolf (2021). Local projections and vars estimate the same impulse responses. *Econometrica 89*(2), 955–980.

Rostagno, M., C. Altavilla, G. Carboni, W. Lemke, R. Motto, and A. Saint-Guilhem (2025). An options-based impact study of the negative interest rate policy and forward guidance. *Journal of monetary economics 152*, 103776–.

Stock, J. H. and M. Watson (2018). Identification and estimation of dynamic causal effects in macroeconomics using external instruments. *Economic Journal 128*(610), 917–948.

Stock, J. H. and M. W. Watson (2003). Forecasting output and inflation: The role of asset prices. *Journal of Economic Literature 41*(3), 788–829.

Tian, Y. S. (2011). Extracting risk-neutral density and its moments from american option prices. *The Journal of Derivatives 18*(3), 17–34.

A Numerical Implementation

This appendix describes the numerical details underlying the estimation procedure in Section 2.1.

Grid construction. The density is estimated on an evenly spaced grid of $n = 300$ log-prices, with date-specific boundaries set at $\log S_0 \pm 2$, where S_0 is the current price of the underlying ETF. For pricing American options via backward induction, I interpolate the coarse-grid nodal probabilities onto a finer grid of 1,500 points using piecewise cubic Hermite interpolation (PCHIP), which preserves monotonicity.

Option selection. No moneyness filter is applied. For each date, all available options with a best bid of at least \$0.50, positive trading volume, and a valid implied volatility are included in the quadratic program. Dates with fewer than five options meeting these criteria are discarded.

Smoothing parameter. The base value of the smoothing parameter is set to $\lambda_0 = 1$. Before entering the iterative procedure, λ is scaled by the number of option contracts m used on that date: $\lambda = \lambda_0 \times m$. This scaling ensures that the smoothness penalty remains balanced against the pricing-error term as the number of options varies across dates.

After the initial scaling, a heuristic diagnostic is applied to verify that the resulting density is not degenerate. Specifically, I compare the cumulative probability mass in the left and right tails of the distribution (at the 16th-percentile node on each side). If the ratio of left-tail to right-tail mass exceeds a large threshold (1000), indicating that the optimizer has concentrated nearly all probability on one side of the grid, λ is increased by 10% and the quadratic program is re-solved. This heuristic adjustment is repeated until the density passes the diagnostic. In practice, this adjustment is triggered only during the extreme market dislocations of March–May 2020, when energy option markets experienced unprecedented volatility following the onset of the COVID-19 pandemic, and those periods are excluded from the main analysis.

Convergence. The iterative early exercise premium procedure is run with a convergence tolerance of $\epsilon = 10^{-5}$ on the absolute change in each element of the early exercise premium vector, and a maximum of 1,000 iterations. To improve convergence stability, the early exercise premium update uses a convex combination of the current and previous estimates in later iterations: for iterations 11 through 20, a fixed mixing weight of $c = 0.3$ is applied

to the previous estimate. For iterations beyond 20, the mixing weight is drawn uniformly at random from $[0, 1]$ at each step. This damping helps avoid fixed-point oscillations that can arise with undamped updates. In practice, the procedure typically converges within 10 iterations.

B Additional Tables

Table 10: XOP Options Data by Year

Year	Observations	Mean Mid Price ($)	Mean Spread ($)	Mean Volume
2008	1,483	13.05	0.77	20.7
2009	2,266	8.08	0.26	104.1
2010	3,616	7.21	0.17	385.5
2011	13,476	10.97	0.18	381.8
2012	14,895	8.43	0.09	372.4
2013	16,453	8.65	0.06	337.9
2014	39,803	12.33	0.32	203.0
2015	55,038	11.54	0.28	198.8
2016	53,063	8.13	0.13	377.6
2017	51,707	6.53	0.07	502.2
2018	59,147	8.03	0.08	446.1
2019	54,801	6.24	0.06	565.4
2020	80,636	4.85	0.20	163.2
2021	99,489	5.67	0.36	104.4

Notes: Yearly averages calculated from daily option metrics.

Table 11: Joint Significance of Risk-Neutral Distribution Moments: Augmented Controls

	IP (1)	PCE (2)	PCEC96 (3)	UNRATE (4)	Core CPI (5)	Energy CPI (6)
Panel A: Pre-COVID sample, augmented controls						
F-statistic	8.37^{***}	0.08	0.82	4.55^{***}	3.76^{**}	2.46^{*}
	(0.000)	(0.971)	(0.485)	(0.005)	(0.013)	(0.067)
Panel B: Full sample, augmented controls						
F-statistic	134.18^{***}	16.35^{***}	8.01^{***}	147.34^{***}	3.14^{**}	3.11^{**}
	(0.000)	(0.000)	(0.000)	(0.000)	(0.028)	(0.029)

Notes. The table reports F-statistics for the joint null hypothesis $H_0 : \beta_1 = \beta_2 = \beta_3 = 0$ from the regressions specified in (9), augmenting the baseline control set with the Gilchrist and Zakrajšek (2012) credit spread index and the excess bond premium. All other specification details are identical to Table 4. p-values in parentheses. Newey-West standard errors with 12 lags. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 12: Out-of-Sample Forecasting: Sensitivity to the Estimation/Evaluation Split

	IP (1)	PCE (2)	PCEC96 (3)	UNRATE (4)	Core CPI (5)	Energy CPI (6)
Panel A: Estimation $= 39$, *Test* $= 70$						
R^2_{OS} (%)	5.56	−31.55	−18.77	−2.51	−4.91	11.91
CW	2.35***	0.20	−1.23	0.24	0.40	2.06**
Panel B: Estimation $= 50$, *Test* $= 59$						
R^2_{OS} (%)	3.52	−20.22	−2.96	−0.54	−3.50	−8.62
CW	1.90**	−0.65	0.23	0.61	−0.07	1.38*
Panel C: Estimation $= 60$, *Test* $= 49$						
R^2_{OS} (%)	1.12	−20.20	−2.24	0.89	−0.59	−11.61
CW	1.74**	−2.01	0.02	0.77	0.45	0.88
Panel D: Estimation $= 70$, *Test* $= 39$ *(baseline)*						
R^2_{OS} (%)	10.46	−10.51	−0.95	10.19	1.44	−2.18
CW	2.43***	−1.58	0.18	2.13**	0.80	1.21

Notes. The table reports out-of-sample R^2_{OS} and the Clark-West MSPE-adjusted statistic for four choices of the estimation/evaluation split. Panel D reproduces the baseline results in Table 5. The full model includes the three lagged RND moments (variance, skewness, kurtosis) in addition to the controls of the AR benchmark. Both the full model and the AR benchmark include four own lags and financial controls (*L*.S&P 500, *L*.VIX, *L*.Oil Price, *L*.OVX). The estimation sample starts in October 2010 and ends in February 2020 across all panels and only the split between estimation and evaluation varies. $R^2_{OS} = 1 - \text{MSFE}_{\text{Full}}/\text{MSFE}_{\text{AR}}$ following Campbell and Thompson (2008). CW is the MSPE-adjusted statistic of Clark and West (2007) (one-sided test); * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ based on standard normal critical values.

Table 13: Out-of-Sample Forecasting with Principal Components: Sensitivity to the Estimation/Evaluation Split

	IP (1)	PCE (2)	PCEC96 (3)	UNRATE (4)	Core CPI (5)	Energy CPI (6)
Panel A: Estimation = 39, Evaluation = 70						
R^2_{OS} (%, 1 PC)	−0.96	−8.37	−1.50	−1.11	−4.21	4.43
R^2_{OS} (%, 2 PCs)	6.32	−16.25	−6.32	−3.01	−4.42	8.58
R^2_{OS} (%, 3 PCs)	4.66	−15.67	−7.02	−2.32	−1.00	2.44
CW (1 PC)	−0.35	−0.46	−0.71	0.11	−0.29	1.39*
CW (2 PCs)	2.16**	−0.62	−1.00	−0.26	−0.30	2.28**
CW (3 PCs)	2.04**	−1.05	−1.13	0.05	0.52	1.71**
Panel B: Estimation = 50, Evaluation = 59						
R^2_{OS} (%, 1 PC)	−0.95	−5.52	0.56	−3.00	−3.56	−8.95
R^2_{OS} (%, 2 PCs)	6.73	−10.71	1.62	−3.21	−3.55	−3.77
R^2_{OS} (%, 3 PCs)	5.66	−11.27	0.60	−2.20	0.15	−5.48
CW (1 PC)	−0.28	−1.20	0.65	−0.43	−0.68	−0.12
CW (2 PCs)	2.07**	−1.07	0.94	−0.28	−0.64	1.64*
CW (3 PCs)	2.06**	−1.29	0.53	0.10	0.53	1.41*
Panel C: Estimation = 60, Evaluation = 49						
R^2_{OS} (%, 1 PC)	−2.47	−4.43	0.54	−4.23	−3.28	−8.71
R^2_{OS} (%, 2 PCs)	1.46	−11.49	1.65	−3.36	−2.73	−7.63
R^2_{OS} (%, 3 PCs)	0.56	−11.74	0.70	−1.82	1.40	−6.19
CW (1 PC)	−1.51	−1.01	0.62	−0.68	−0.57	−0.39
CW (2 PCs)	1.34	−1.70	0.89	−0.27	−0.39	1.04
CW (3 PCs)	1.42*	−1.82	0.51	0.19	0.75	1.09
Panel D: Estimation = 70, Evaluation = 39 (baseline)						
R^2_{OS} (%, 1 PC)	−1.15	−3.67	0.09	−1.09	−3.15	−6.69
R^2_{OS} (%, 2 PCs)	8.69	−5.95	2.20	1.85	−1.93	−6.28
R^2_{OS} (%, 3 PCs)	8.88	−6.55	1.37	4.87	2.98	−4.30
CW (1 PC)	−0.76	−0.72	0.25	0.14	−0.47	−0.16
CW (2 PCs)	2.26**	−1.09	0.93	0.96	−0.13	0.74
CW (3 PCs)	2.32**	−1.20	0.61	1.58*	1.03	0.87

Notes. The table reports out-of-sample R^2_{OS} and the Clark-West MSPE-adjusted statistic for four choices of the estimation/evaluation split, under one-, two-, and three-principal-components specifications. Panel D reproduces the corresponding rows of Table 7. The full model includes the first k principal components extracted from 15 lagged RND features (ten standardized moments and five percentiles) in addition to the controls of the AR benchmark. Both the PCA loadings and the regression coefficients are re-estimated at each step of the expanding window to avoid look-ahead bias. All panels use an estimation sample running from October 2010 to February 2020 and only the split between estimation and evaluation varies. $R^2_{OS} = 1 - \text{MSFE}_{\text{Full}}/\text{MSFE}_{\text{AR}}$ following Campbell and Thompson (2008). CW is the MSPE-adjusted statistic of Clark and West (2007) (one-sided test); * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.